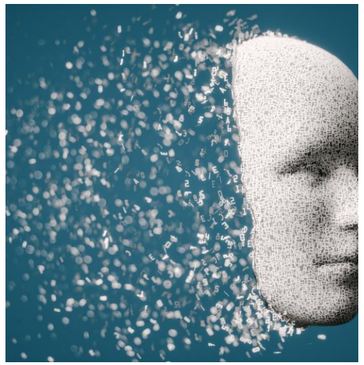
DEPARTEMEN
TEKNIK INFORMATIKA



EA234406 - Pengolahan Citra Digital

Prodi Sarjana Rekayasa Kecerdasan Artifisial

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Rencana Pembelajaran Minggu 1 – 5 PCD

Minggu 1 Pendahuluan dan Dasar-dasar Pengolahan Citra

Minggu 2 Perbaikan Citra Domain Spasial

Minggu 3 Demo Tugas 1 dan Quiz (Online/Offline)

Minggu 4 Perbaikan Citra Domain Frekuensi

Minggu 5 Demo Tugas 2 dan Quiz (Online/Offline)

Pengenalan Citra Digital

"One picture is worth more than ten thousand words."

Anonymous

Apa itu Pengolahan Citra Digital?

Citra = fungsi dua dimensi, f(x, y), di mana x dan y adalah koordinat spasial (bidang), dan amplitudo f pada pasangan koordinat (x, y) disebut intensitas atau tingkat abu-abu gambar pada titik itu.

Ketika x, y, dan nilai intensitas f semuanya terbatas dan besaran diskrit, kita menyebut citra tsb sebagai citra digital.

Pengolahan citra digital = pengolahan citra digital melalui komputer digital.

Citra digital terdiri dari sejumlah elemen yang terbatas, yang masing-masing memiliki lokasi dan nilai tertentu.

Elemen-elemen ini disebut elemen citra atau piksel. Piksel adalah istilah yang paling banyak digunakan untuk menunjukkan elemen citra digital.

Sejarah

Salah satu aplikasi paling awal dari citra digital adalah di industri surat kabar, ketika citra pertama kali dikirim dengan kabel bawah laut antara London dan New York.

Sistem transmisi citra Bartlane pada awal 1920-an mengurangi waktu yang diperlukan untuk mengirim citra melintasi Atlantik dari lebih dari seminggu menjadi kurang dari tiga jam.

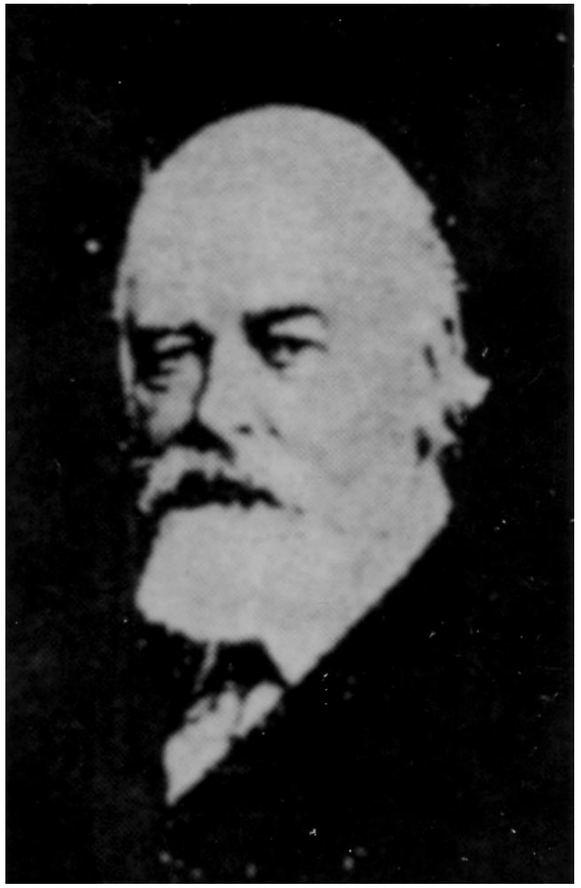
Printer khusus mengkodekan gambar untuk transmisi kabel, kemudian merekonstruksinya di ujung penerima.



Sejarah

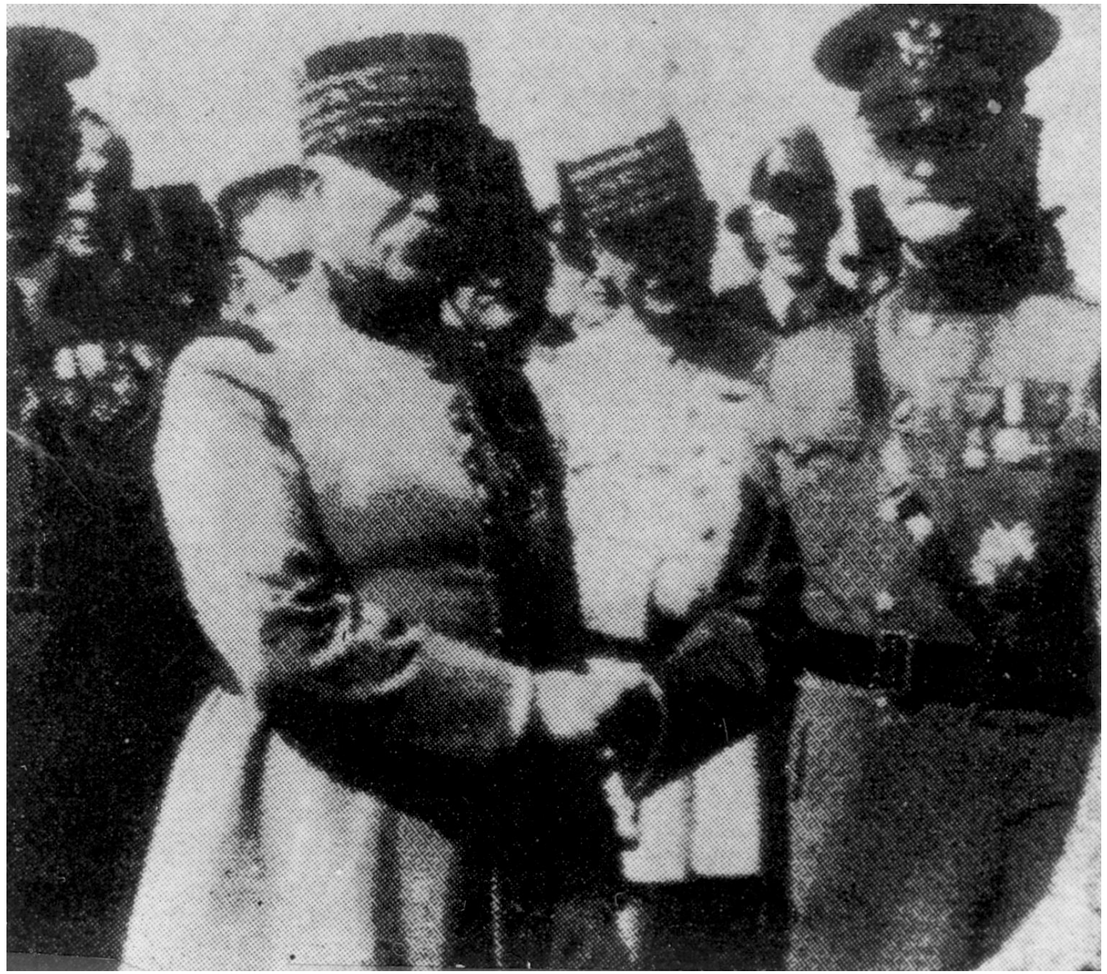
Metode printing yang digunakan sebelumnya ditinggalkan menjelang akhir tahun 1921 demi teknik berdasarkan reproduksi fotografi yang dibuat dari pita yang dilubangi di terminal penerima telegraf.

Peningkatan dibandingkan citra sebelumnya terbukti, baik dalam kualitas nada maupun dalam hal resolusi.



Sejarah

The early Bartlane systems produced images having 15 levels of gray →

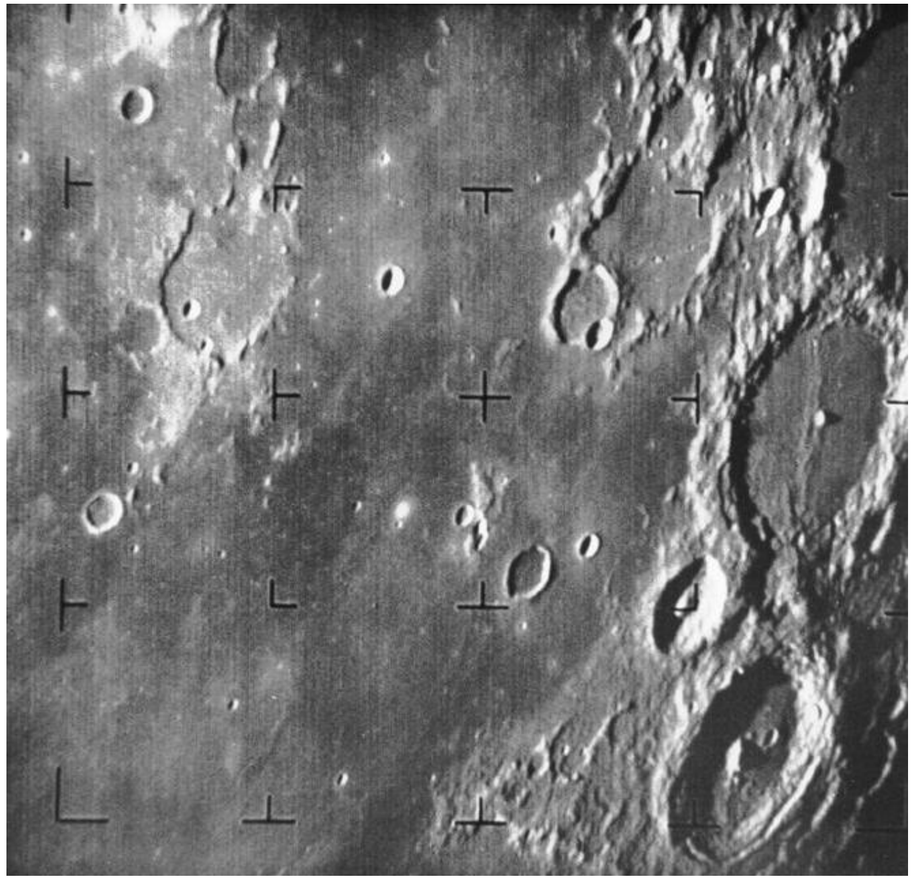


Sejarah

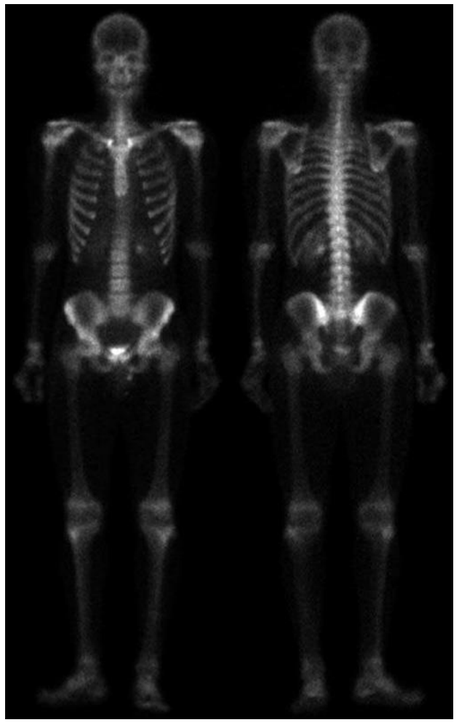
Komputer pertama yang cukup kuat untuk melakukan tugas pengolahan citra digital yang bermakna muncul pada awal 1960-an.

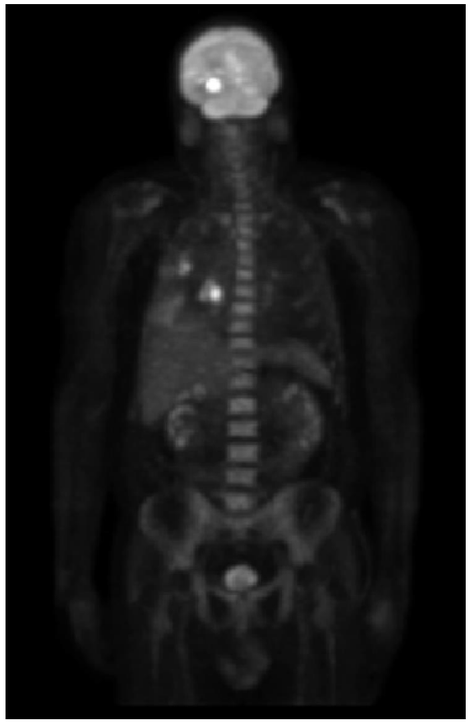
PCD digunakan untuk meningkatkan citra dari luar angkasa yang dimulai pada tahun 1964 di Jet Propulsion Laboratory (Pasadena, California). ketika gambar bulan yang ditransmisikan oleh Ranger 7 diproses oleh komputer untuk mengoreksi berbagai jenis distorsi gambar yang melekat pada kamera televisi on-board.

Citra pertama bulan yang diambil oleh pesawat ruang angkasa AS →

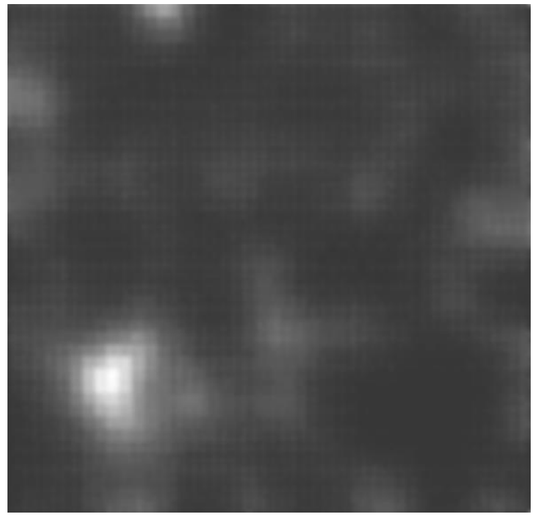


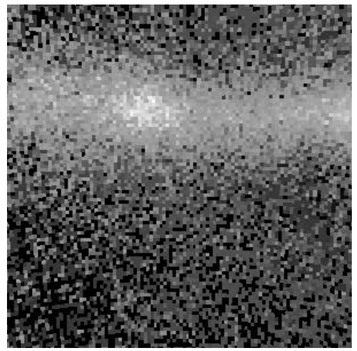
Aplikasi PCD – Sinar Gamma





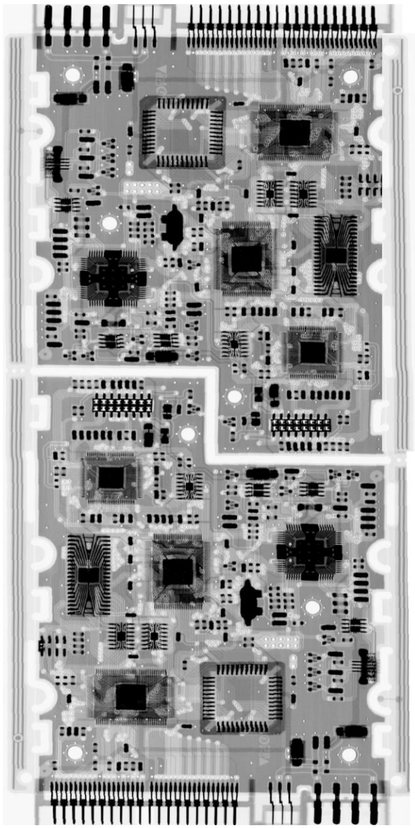
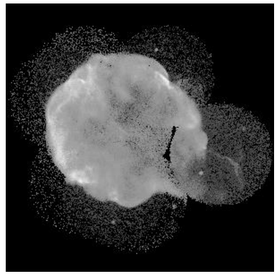
Examples of gamma-ray imaging.
(a) Bone scan.
(b) PET image.
(c) Cygnus Loop.
(d) Gamma radiation (bright spot) from a reactor valve.





Aplikasi PCD – Sinar X



Examples of
X-ray imaging.
(a) Chest X-ray.
(b) Aortic
angiogram.
(c) Head CT.
(d) Circuit boards.
(e) Cygnus Loop.

Aplikasi PCD – remote sensing

Band No.	Name	Wavelength (μm)	Characteristics and Uses
1	Visible blue	0.45–0.52	Maximum water penetration
2	Visible green	0.53–0.61	Measures plant vigor
3	Visible red	0.63–0.69	Vegetation discrimination
4	Near infrared	0.78–0.90	Biomass and shoreline mapping
5	Middle infrared	1.55–1.75	Moisture content: soil/vegetation
6	Thermal infrared	10.4–12.5	Soil moisture; thermal mapping
7	Short-wave infrared	2.09–2.35	Mineral mapping

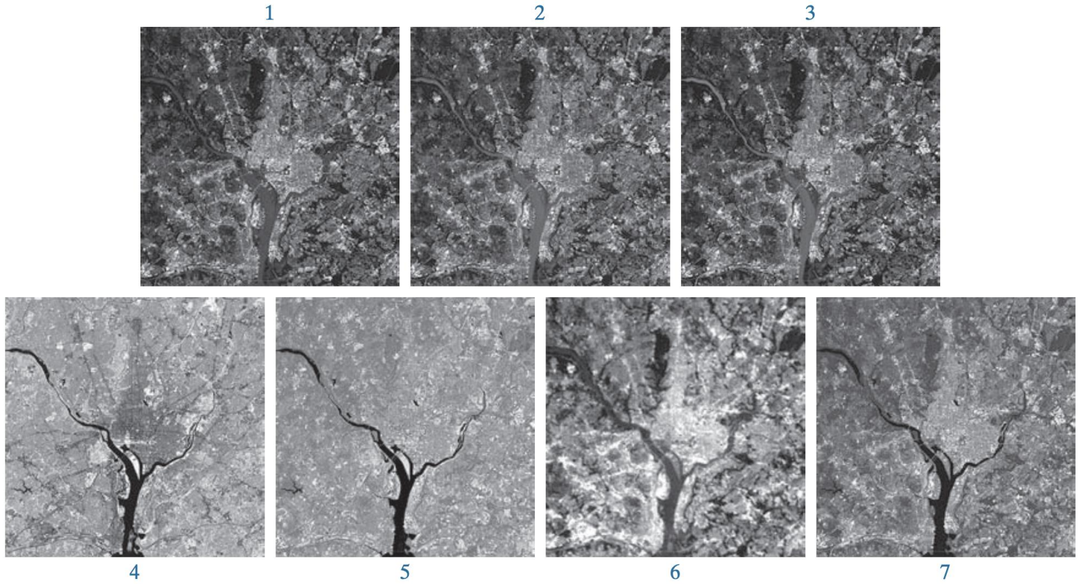
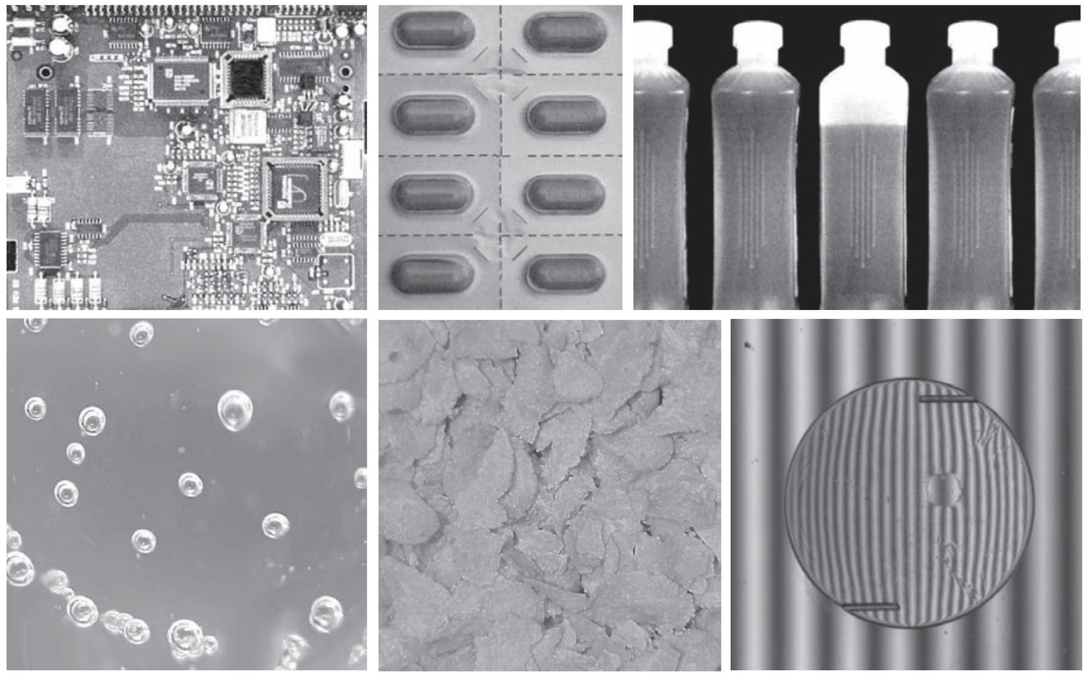


FIGURE 1.10 LANDSAT satellite images of the Washington, D.C. area. The numbers refer to the thematic bands in Table 1.1. (Images courtesy of NASA.)

Aplikasi PCD – spektrum cahaya



a b c
d e f

FIGURE 1.14 Some examples of manufactured goods checked using digital image processing. (a) Circuit board controller. (b) Packaged pills. (c) Bottles. (d) Air bubbles in a clear plastic product. (e) Cereal. (f) Image of intraocular implant. (Figure (f) courtesy of Mr. Pete Sites, Perceptics Corporation.)

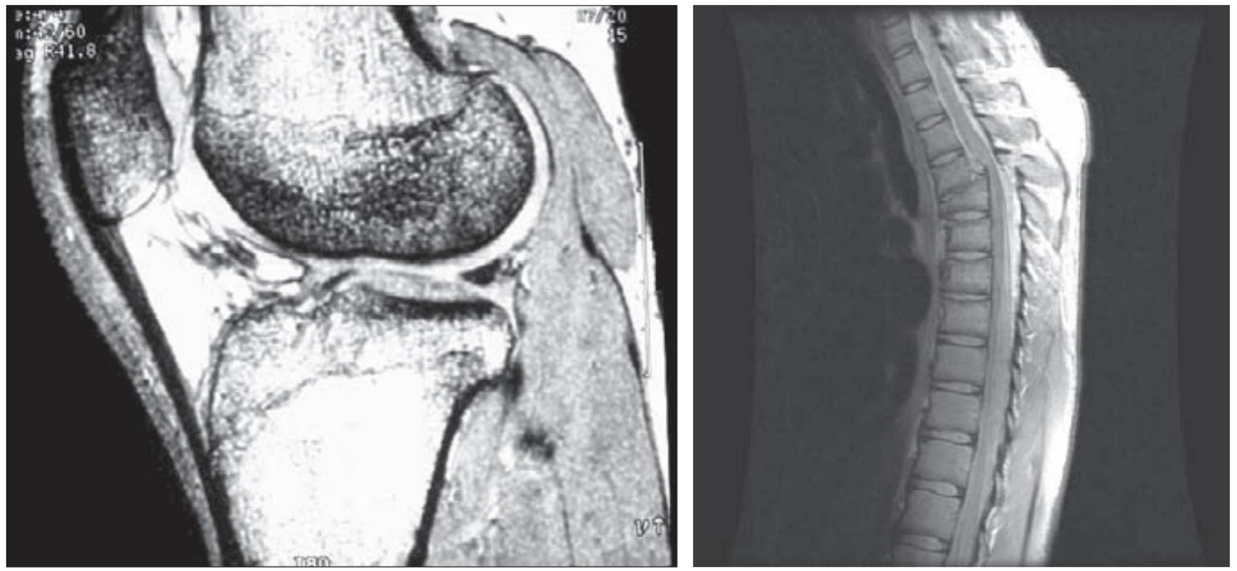
Aplikasi PCD – spektrum cahaya

a b
c
d

FIGURE 1.15
Some additional examples of imaging in the visible spectrum. (a) Thumb print. (b) Paper currency. (c) and (d) Automated license plate reading. (Figure (a) courtesy of the National Institute of Standards and Technology. Figures (c) and (d) courtesy of Dr. Juan Herrera, Perceptics Corporation.)



Aplikasi PCD – gelombang radio



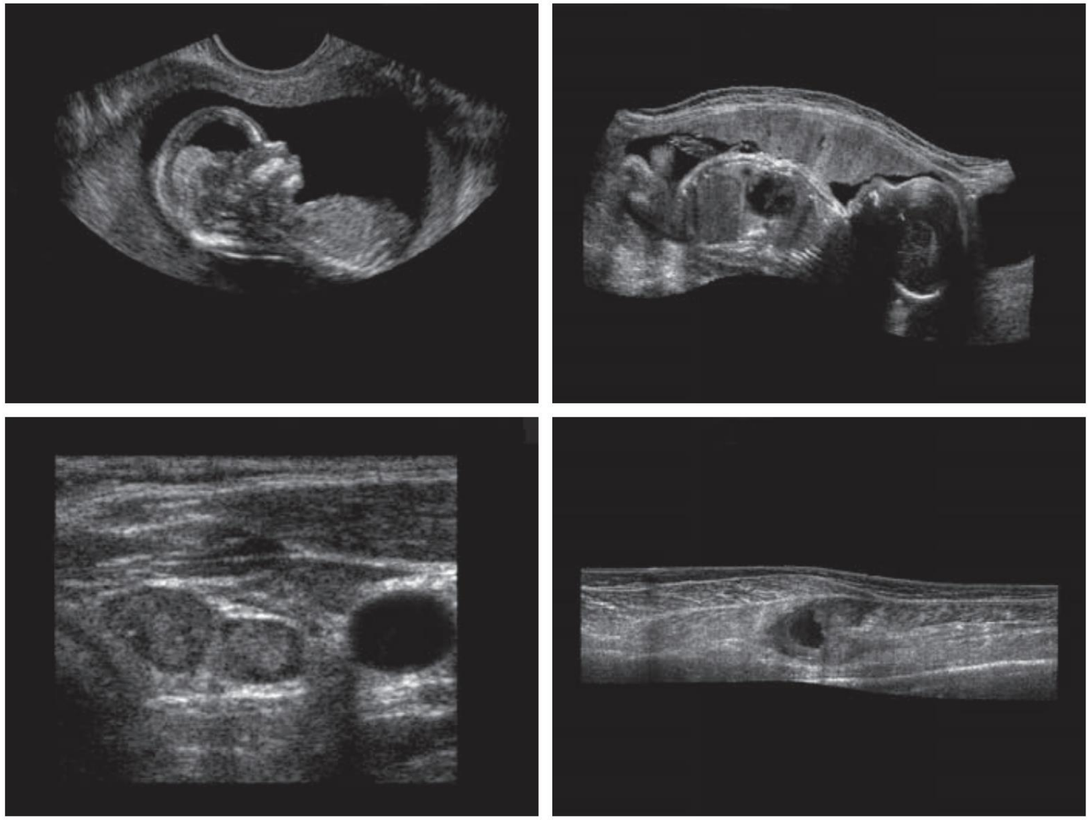
a b

FIGURE 1.17 MRI images of a human (a) knee, and (b) spine. (Figure (a) courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, and (b) courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

Aplikasi PCD – ultrasound

a b
c d

FIGURE 1.20
Examples of ultrasound imaging. (a) A fetus. (b) Another view of the fetus. (c) Thyroids. (d) Muscle layers showing lesion. (Courtesy of Siemens Medical Systems, Inc., Ultrasound Group.)



Pengolahan Citra Digital vs Visi Komputer

Tujuan keduanya memiliki perbedaan.

Pengolahan citra digital fokus pada meningkatkan kualitas citra, menyesuaikan kecerahan, menajamkan detail, mengurangi noise/derau.

Visi komputer memiliki tujuan lebih jauh seperti mengenali dan menginterpretasikan citra atau video seperti manusia mengenali citra/video. Contoh aplikasi visi komputer: pengenalan wajah, deteksi objek, dan analisis video secara realtime.

Di industri, pasar pemrosesan citra secara global akan meningkat senilai $378.71 miliar pada tahun 2034. sedangkan pasar visi komputer meningkat menjadi $111.43 miliar pada tahun 2033.

FEATURES	COMPUTER VISION	IMAGE PROCESSING
PURPOSE	Interprets and understands images	Enhances or modifies images
FOCUS	Extracts meaning and insights	Improves quality and features
TECHNIQUES	Object detection, segmentation, pose estimation	Noise reduction, edge detection, segmentation
END GOAL	Decision-making and automation	Prepares images for analysis

Sumber: ultralytics.com

Pemrosesan Citra dan Visi Komputer

Keduanya dapat digunakan secara bersamaan. Pemrosesan citra digunakan untuk meningkatkan kualitas citra dan mengekstraksi fitur utama, dan visi komputer melanjutkan dengan menyediakan simpulan analisis citra.

Keduanya akan dapat meningkatkan keakuratan, efisiensi, dan kemampuan analisis data visual pada aplikasi nyata.

Dasar-dasar Pengolahan Citra Digital

CITRA DIGITAL

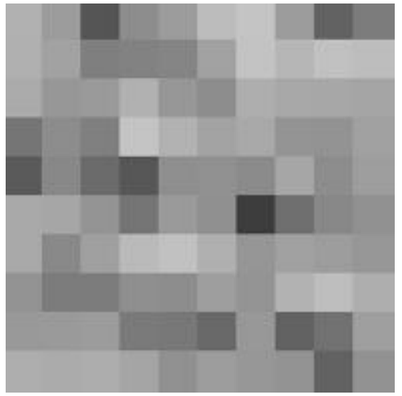


$F(x, y)$

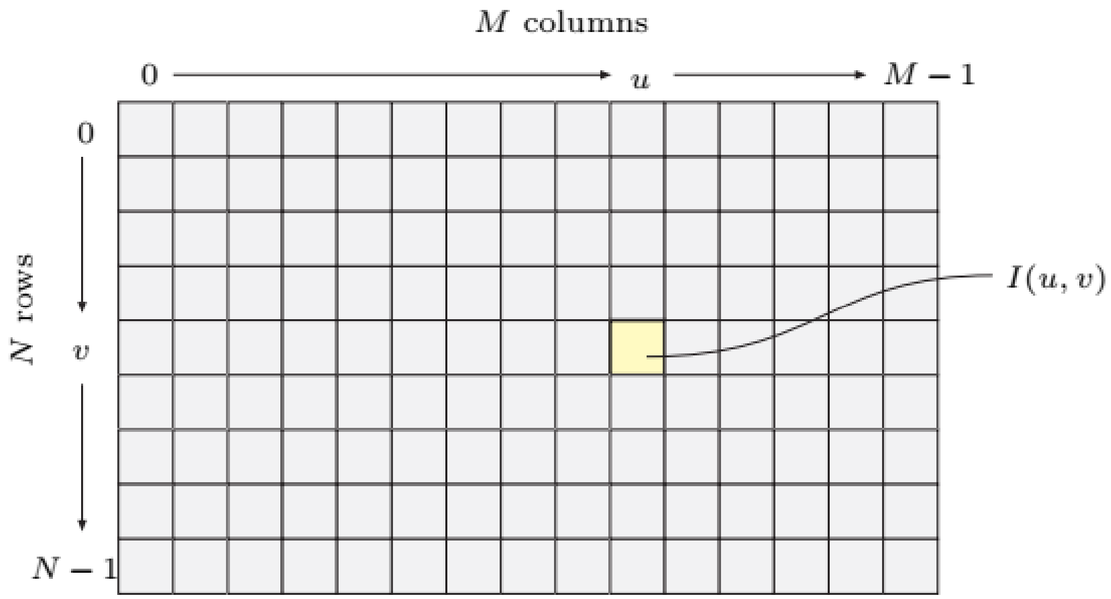
$\longrightarrow$

$$
\begin{array}{cccccccccc}
148 & 123 & 52 & 107 & 123 & 162 & 172 & 123 & 64 & 89 \cdots \\
147 & 130 & 92 & 95 & 98 & 130 & 171 & 155 & 169 & 163 \cdots \\
141 & 118 & 121 & 148 & 117 & 107 & 144 & 137 & 136 & 134 \cdots \\
82 & 106 & 93 & 172 & 149 & 131 & 138 & 114 & 113 & 129 \cdots \\
57 & 101 & 72 & 54 & 109 & 111 & 104 & 135 & 106 & 125 \cdots \\
138 & 135 & 114 & 82 & 121 & 110 & 34 & 76 & 101 & 111 \cdots \\
138 & 102 & 128 & 159 & 168 & 147 & 116 & 129 & 124 & 117 \cdots \\
113 & 89 & 89 & 109 & 106 & 126 & 114 & 150 & 164 & 145 \cdots \\
120 & 121 & 123 & 87 & 85 & 70 & 119 & 64 & 79 & 127 \cdots \\
145 & 141 & 143 & 134 & 111 & 124 & 117 & 113 & 64 & 112 \cdots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots
\end{array}
$$

$I(u, v)$



KOORDINAT CITRA DIGITAL



RESOLUSI CITRA (SPASIAL DAN INTENSITAS)

- Resolusi spasial direpresentasikan dengan banyaknya titik (piksel) pada satu satuan jarak, misalnya *dots per inch* (dpi)
  - dpi digunakan di industri percetakan di US
    - Koran dicetak dengan resolusi 75 dpi
    - Majalah: 133 dpi
    - Brosur: 175 dpi
    - Buku: 2400 dpi

RESOLUSI SPASIAL

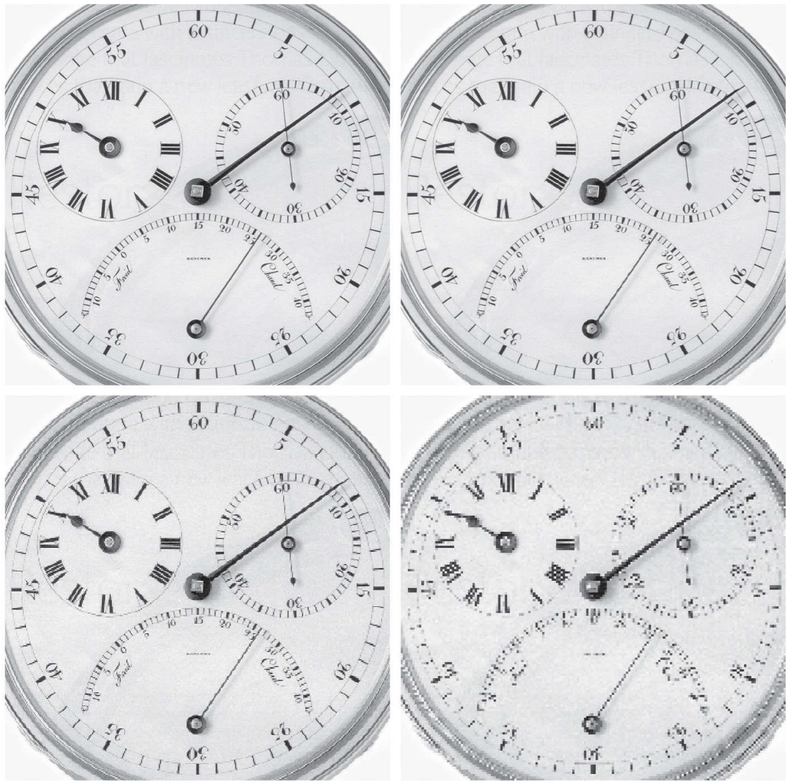
(a) 930 dpi

(b) 300 dpi

(c) 150 dpi

(d) 72 dpi

a b
c d



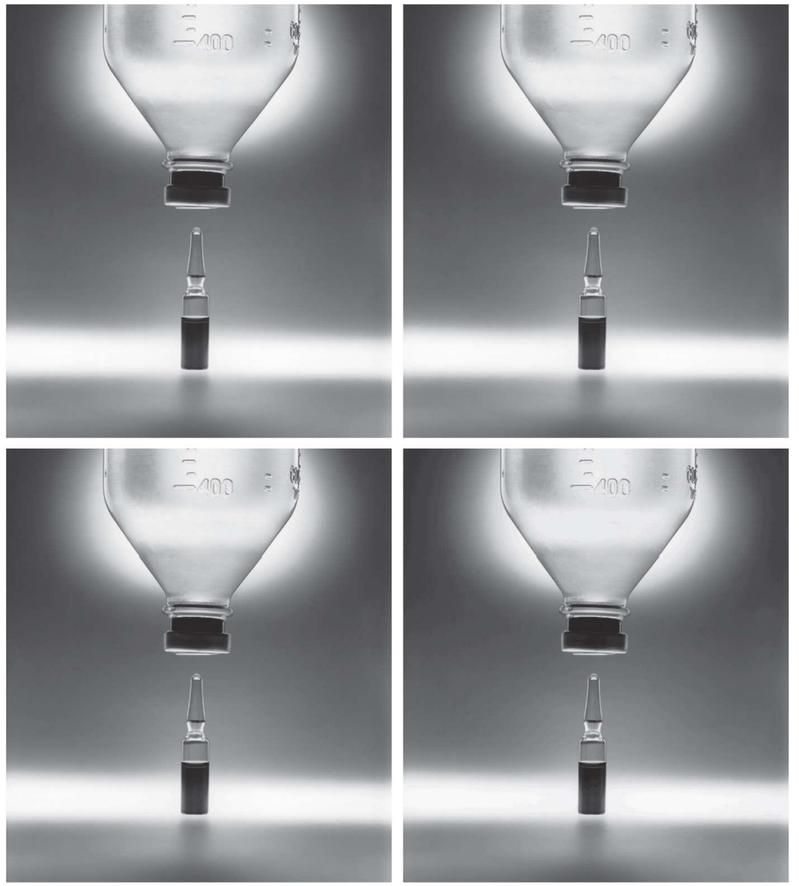
RESOLUSI INTENSITAS

Karena pertimbangan hardware, level intensitas umumnya bilangan pangkat 2, misalnya 8 bits

Resolusi intensitas 8 bits = 256 level intensitas

RESOLUSI INTENSITAS

a b
c d



(a) 256-level image

(b) 128-level

(c) 64-level

(d) 32-level

RESOLUSI INTENSITAS

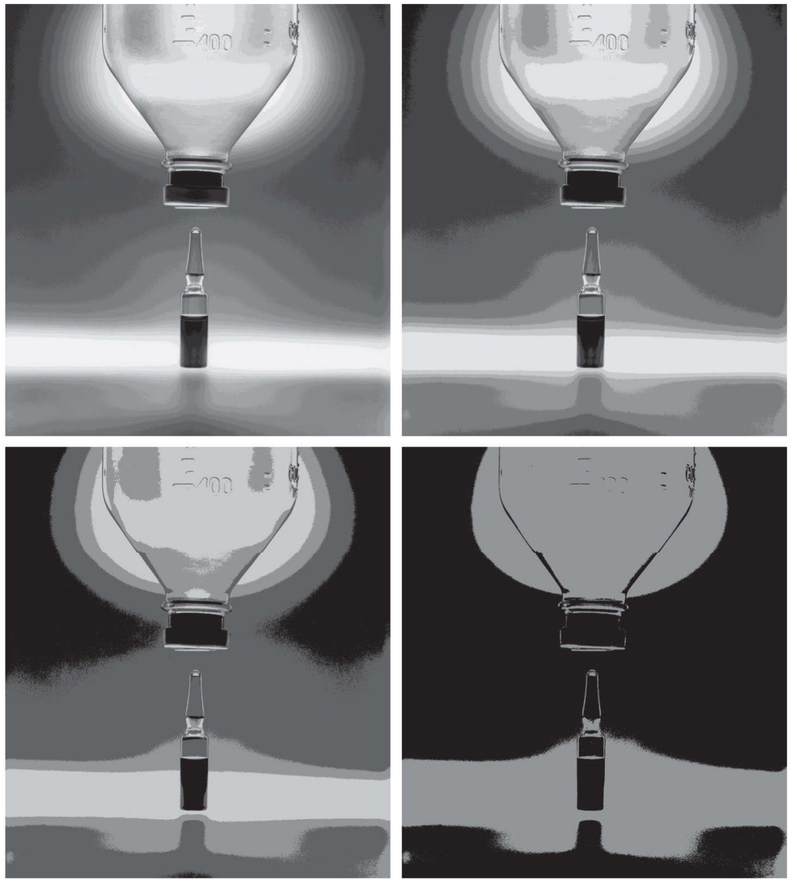
e. 16-level image

f. 8-level

g. 4-level

h. 2-level

e f
g h



BIT DEPTHS OF COMMON IMAGE TYPES AND TYPICAL APPLICATION DOMAINS

Grayscale (Intensity Images):

Chan.	*Bits/Pix.*	*Range*	*Use*
1	1	$[0, 1]$	Binary image: document, illustration, fax
1	8	$[0, 255]$	Universal: photo, scan, print
1	12	$[0, 4095]$	High quality: photo, scan, print
1	14	$[0, 16383]$	Professional: photo, scan, print
1	16	$[0, 65535]$	Highest quality: medicine, astronomy

Color Images:

Chan.	*Bits/Pix.*	*Range*	*Use*
3	24	$[0, 255]^3$	RGB, universal: photo, scan, print
3	36	$[0, 4095]^3$	RGB, high quality: photo, scan, print
3	42	$[0, 16383]^3$	RGB, professional: photo, scan, print
4	32	$[0, 255]^4$	CMYK, digital prepress

Special Images:

Chan.	*Bits/Pix.*	*Range*	*Use*
1	16	$[-32768, 32767]$	Integer values pos./neg., increased range
1	32	$\pm 3.4 \cdot 10^{38}$	Floating-point values: medicine, astronomy
1	64	$\pm 1.8 \cdot 10^{308}$	Floating-point values: internal processing

NEIGHBORS OF PIXELS

A pixel p at coordinates (x, y) has two horizontal and two vertical neighbors with coordinates

$$(x+1, y), (x-1, y), (x, y+1), (x, y-1)$$

This set of pixels, called the 4-*neighbors* of p, is denoted $N_4(p)$.

The four *diagonal* neighbors of p have coordinates

$$(x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)$$

and are denoted $N_D(p)$. These neighbors, together with the 4-neighbors, are called the 8-*neighbors* of p, denoted by $N_8(p)$. The set of image locations of the neighbors of a point p is called the *neighborhood* of p. The neighborhood is said to be *closed* if it contains p. Otherwise, the neighborhood is said to be *open*.

ELEMENTWISE VS MATRIX OPS

An *elementwise operation* involving one or more images is carried out on a *pixel-by-pixel* basis. We mentioned earlier in this chapter that images can be viewed equivalently as matrices. In fact, as you will see later in this section, there are many situations in which operations between images are carried out using matrix theory. It is for this reason that a clear distinction must be made between elementwise and matrix operations. For example, consider the following 2×2 images (matrices):

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

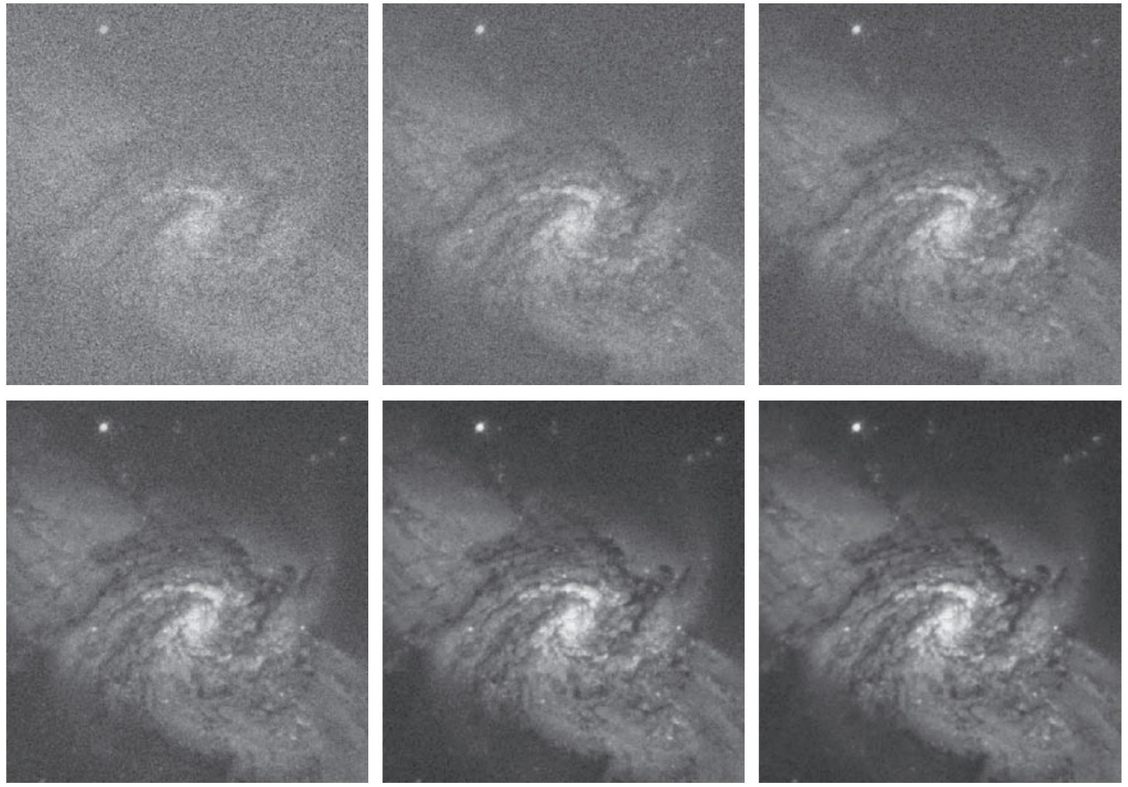
The *elementwise product* (often denoted using the symbol $\odot$ or $\otimes$) of these two images is

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \odot \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

That is, the elementwise product is obtained by multiplying pairs of *corresponding* pixels. On the other hand, the *matrix product* of the images is formed using the rules of matrix multiplication:

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

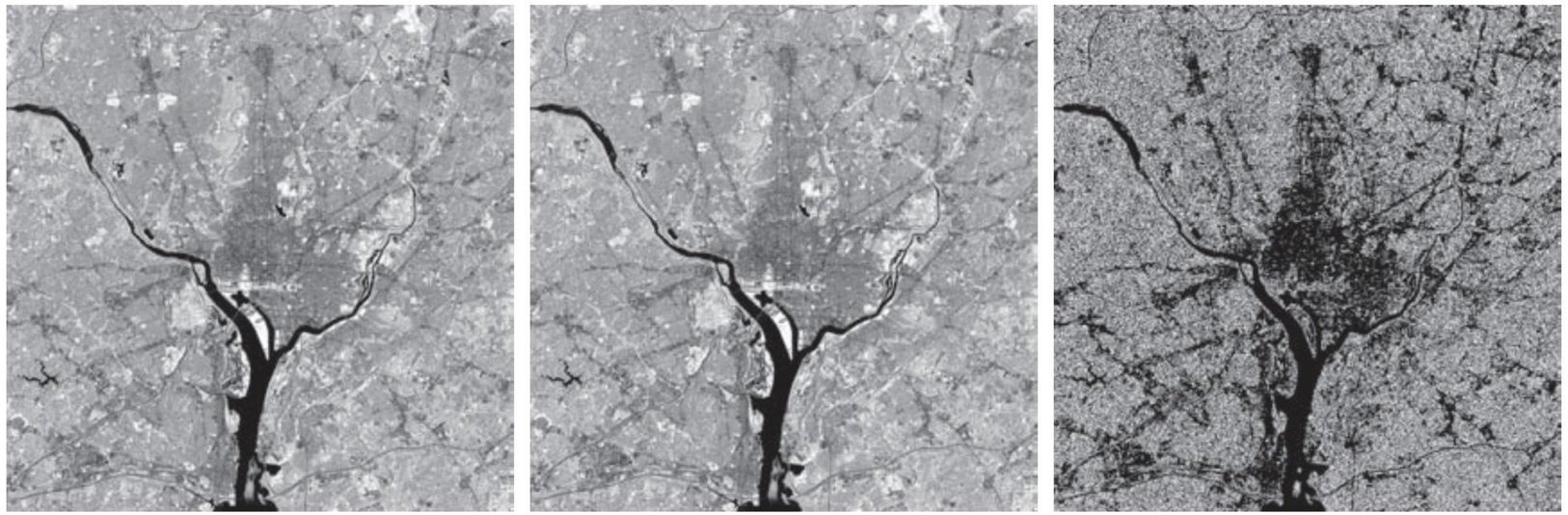
USING IMAGE ADDITION (AVERAGING) FOR NOISE REDUCTION



a b c
d e f

FIGURE 2.29 (a) Image of Galaxy Pair NGC 3314 corrupted by additive Gaussian noise. (b)-(f) Result of averaging 5, 10, 20, 50, and 1,00 noisy images, respectively. All images are of size 566×598 pixels, and all were scaled so that their intensities would span the full [0, 255] intensity scale. (Original image courtesy of NASA.)

COMPARING IMAGES USING SUBTRACTION



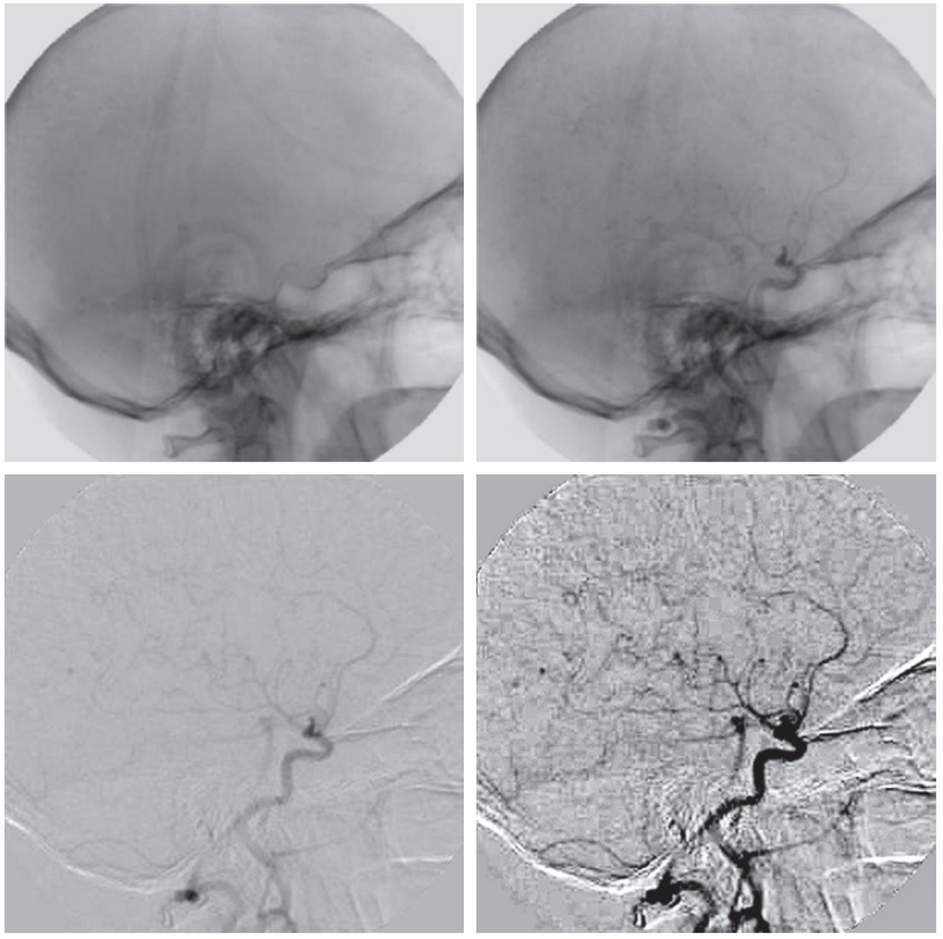
a b c

FIGURE 2.30 (a) Infrared image of the Washington, D.C. area. (b) Image resulting from setting to zero the least significant bit of every pixel in (a). (c) Difference of the two images, scaled to the range [0, 255] for clarity. (Original image courtesy of NASA.)

COMPARING IMAGES USING SUBTRACTION

a b
c d

FIGURE 2.32
Digital subtraction angiography. (a) Mask image. (b) A live image. (c) Difference between (a) and (b). (d) Enhanced difference image. (Figures (a) and (b) courtesy of the Image Sciences Institute, University Medical Center, Utrecht, The Netherlands.)



USING IMAGE MULTIPLICATION AND DIVISION FOR SHADING CORRECTION AND FOR MASKING



a b c

FIGURE 2.33 Shading correction. (a) Shaded test pattern. (b) Estimated shading pattern. (c) Product of (a) by the reciprocal of (b). (See Section 3.5 for a discussion of how (b) was estimated.)

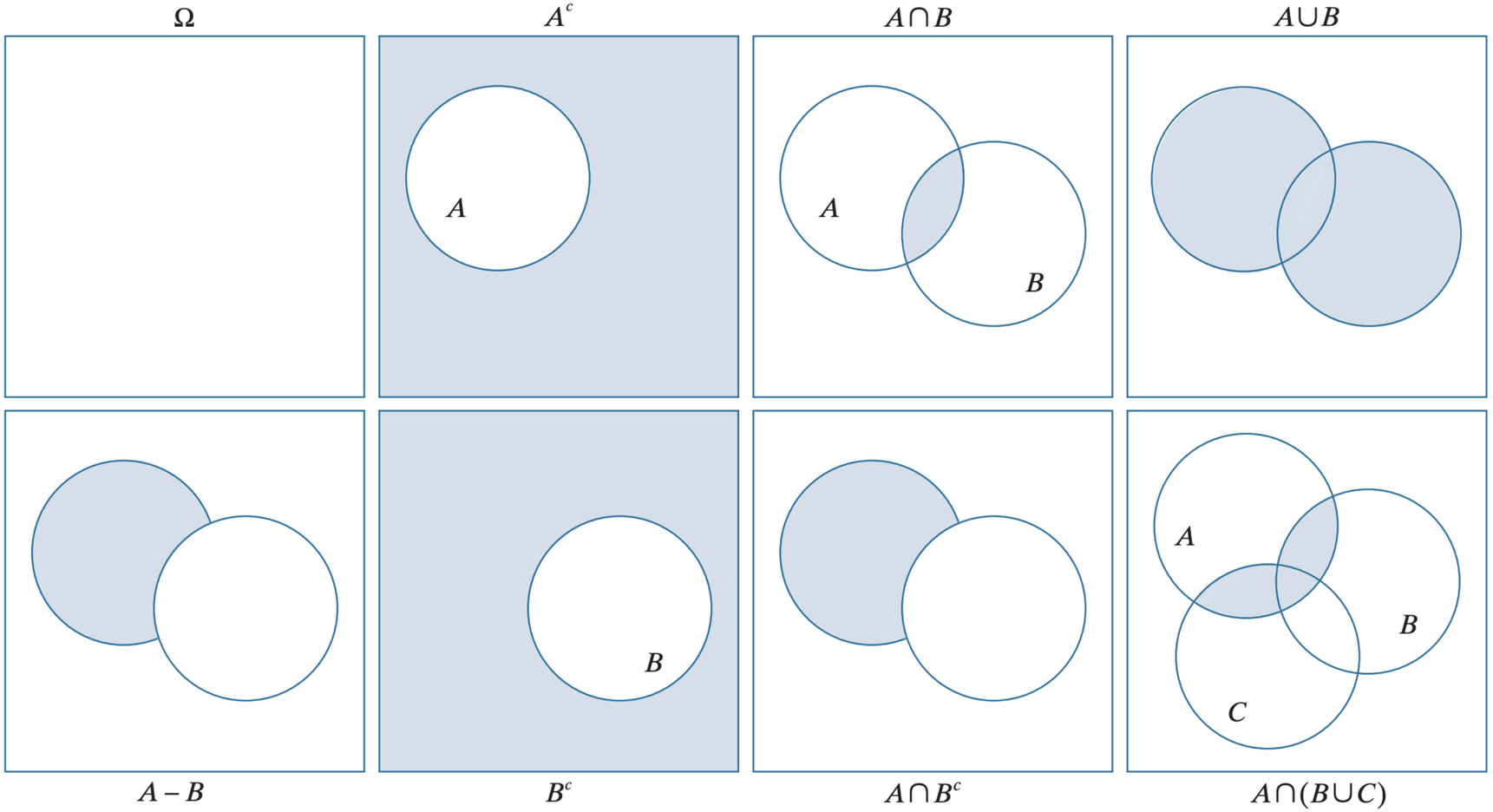
USING IMAGE MULTIPLICATION AND DIVISION FOR SHADING CORRECTION AND FOR MASKING



a b c

FIGURE 2.34 (a) Digital dental X-ray image. (b) ROI mask for isolating teeth with fillings (white corresponds to 1 and black corresponds to 0). (c) Product of (a) and (b).

BASIC SET OPERATIONS



SPATIAL OPERATIONS

three broad categories:

(1) single-pixel operations,

(2) neighborhood operations, and

(3) geometric spatial transformations.

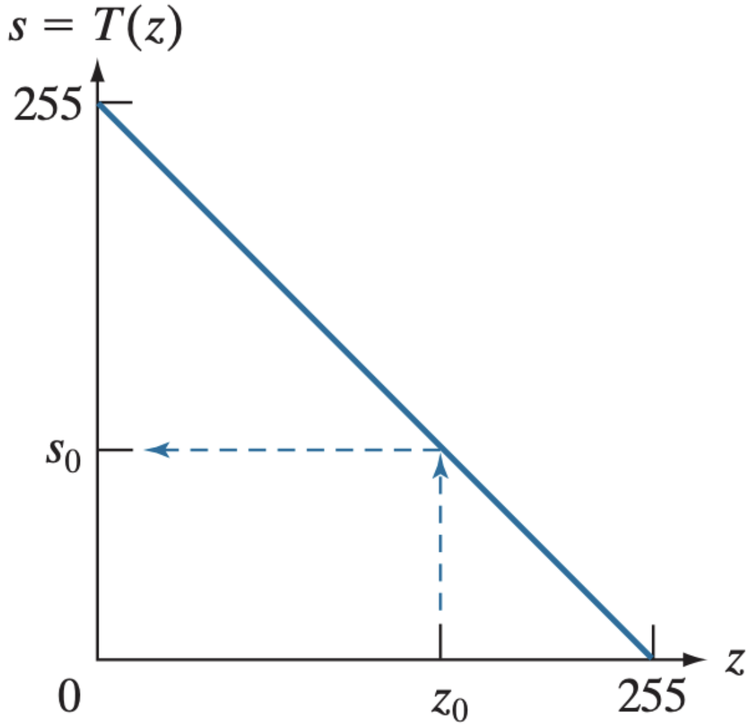
SINGLE-PIXEL OPERATIONS

Alter the intensity of its pixels individually using a transformation function, T:

$s = T(z)$

where z is the intensity of a pixel in the original image and s is the (mapped) intensity of the corresponding pixel in the processed image.

SINGLE-PIXEL OPERATIONS



Apa yang dilakukan fungsi transformasi intensitas disamping?

NEIGHBORHOOD OPERATIONS

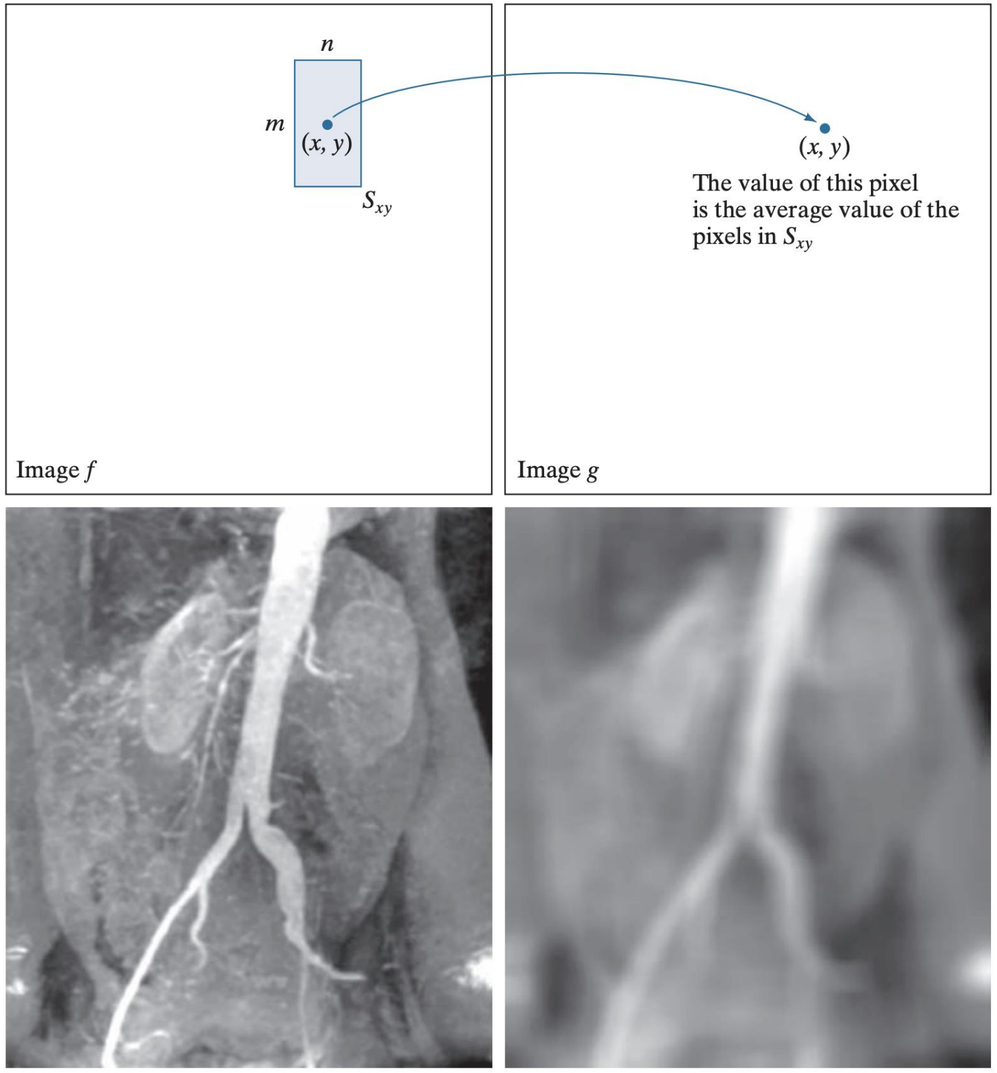
Neighborhood processing generates a corresponding pixel at the same coordinates in an output (processed) image, such that the value of that pixel is determined by a specified operation on the neighborhood of pixels in the input image.

For example, suppose that the specified operation is to compute the average value of the pixels in a rectangular neighborhood.

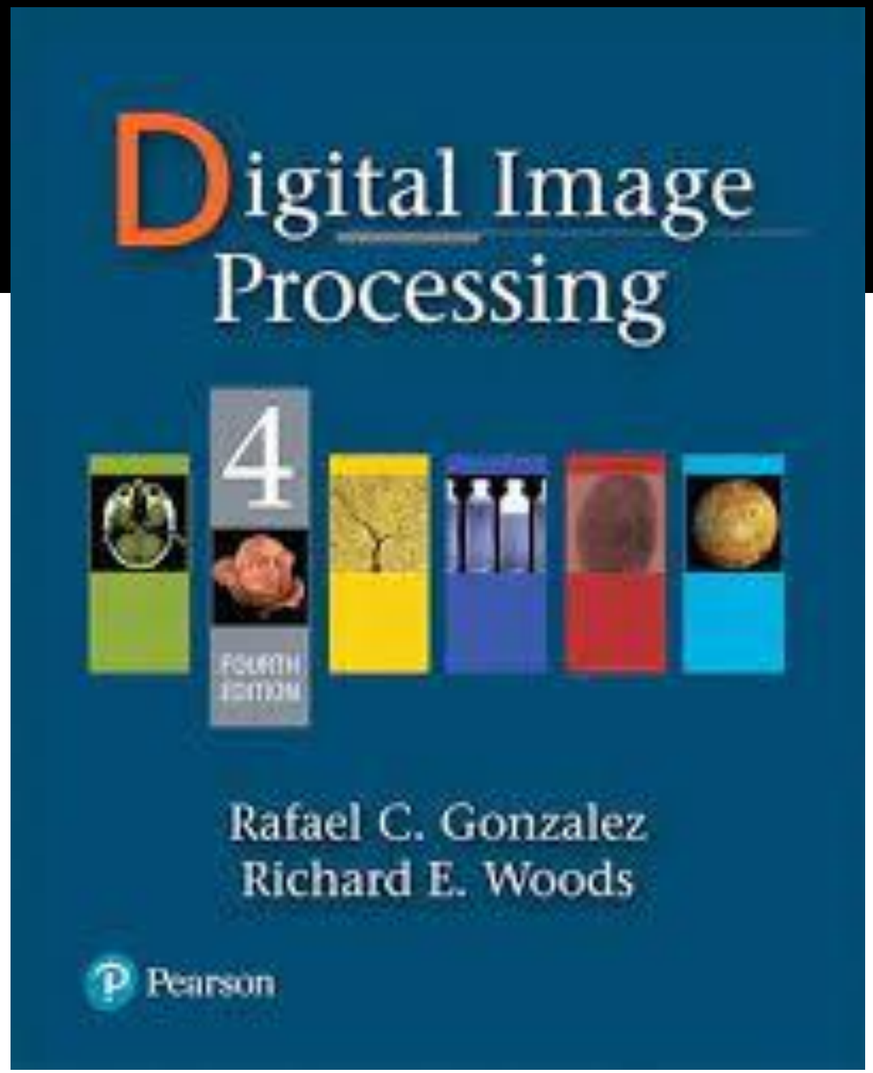
NEIGHBORHOOD OPERATIONS

a b
c d

FIGURE 2.39
Local averaging using neighborhood processing. The procedure is illustrated in (a) and (b) for a rectangular neighborhood. (c) An aortic angiogram (see Section 1.3). (d) The result of using Eq. (2-43) with $m = n = 41$. The images are of size 790×686 pixels. (Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)



Fundamental steps in digital image processing (Gonzales & Woods, 2018).



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